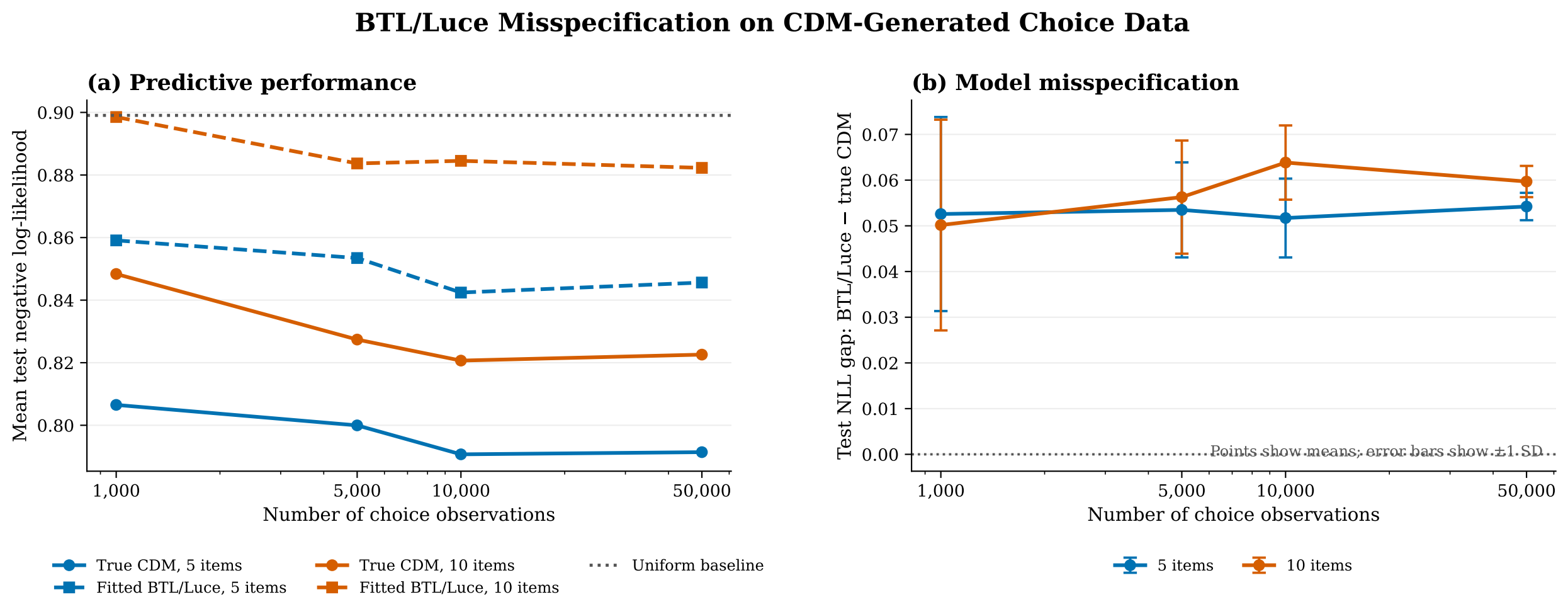
BTL/Luce Misspecification on CDM-Generated Choice Data
(a) Predictive performance
Mean test negative log-likelihood
0.90
0.88
0.86
0.84
0.82
0.80
1,000
5,000
10,000
50,000
Number of choice observations
True CDM, 5 items
True CDM, 10 items
Uniform baseline
Fitted BTL/Luce, 5 items
Fitted BTL/Luce, 10 items
(b) Model misspecification
Test NLL gap: BTL/Luce − true CDM
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
Points show means; error bars show ±1 SD
1,000
5,000
10,000
50,000
Number of choice observations
5 items
10 items